

Power Set: Finding All Subsets of a Set

The *power set* of a set is the set of every possible subset of that set. For example, the power set of $\{A, B, C\}$ is $\{\{\}, \{A\}, \{B\}, \{C\}, \{A, B\}, \{A, C\}, \{B, C\}, \{A, B, C\}\}$. This is equivalent to the set of every possible k -combination of a set. After all, the power set of $\{A, B, C\}$ contains all its 0-combinations, 1-combinations, 2-combinations, and 3-combinations.

If you're looking for a real-world example in which you would need to generate the power set of a set, imagine a job interviewer asked you to generate the power set of a set. It is astronomically unlikely you'll need to generate the power set of a set for any other reason, including the job you are interviewing for.

To find every power set of a set, we could reuse our existing `getCombos()` function, calling it repeatedly with each possible k argument. This approach is taken by the `powerSetCombinations.py` and `powerSetCombinations.html` programs in the downloadable resources file from <https://nostarch.com/recursive-book-recursion>.

However, we can use a more efficient way to generate power sets. Let's consider the power set of $\{A, B\}$, which is $\{\{A, B\}, \{A\}, \{B\}, \{\}\}$. Now say we add one more element, C , to the set and want to generate the power set of $\{A, B, C\}$. We have the four sets in the power set of $\{A, B\}$ we already generated; in addition, we have these same four sets but with the element C added to them: $\{\{A, B, C\}, \{A, C\}, \{B, C\}, \{C\}\}$. [Table 6-3](#) shows the pattern of how adding more elements to a set adds more sets to its power set.